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## ME 527 : Biofluid Dynamics

Course Code : ME 527

Course Name : Biofluid Dynamics

L-P-T-C: 3-0-0-3

Intended for: UG final year, and PG students

Prerequisites: ME210 or Equivalent Fluid Mechanics Course

Mutual Exclusion: None

Approval : 57th BoA

### Course Contents

- **Review of Basic Fluid Mechanics:** Fluid Properties; Pressure and Buoyancy; Flow Phenomena (Laminar flow and turbulent flow; Equation for mass, momentum, and energy conservation in incompressible, frictionless fluids); Viscosity and Resistance (Poiseuille's law for laminar flow resistance in tubes) (6 Hours)
- **Biorheology:** Newtonian and non-Newtonian fluids. Understanding how these macromolecules respond to mechanical forces. Investigating the flow properties of large biological molecules. (6 Hours)
- **Hemodynamics:** Circulation in the heart, blood and lymphatic vessels. Blood properties. Steady flow in tubes. Pulsatile flow in a rigid tube. Pulsatile flow in an elastic tube. Blood flow dynamics in arteries and veins. Microcirculation. Flow in specific vessels and arteries. Heart-valve hemodynamics. Diseases related to obstruction of blood flow. (10 Hours)
- **Respiratory Biofluid Dynamics:** Flow in the upper and lower airways. Alveolar ventilation. Airflow in the lungs. Mechanics of breathing. Gas exchange and transport. (6 Hours)
- **Cardiovascular Implants & Drug Delivery Systems:** Function of artificial heart valves and devices (pressure distribution and fluid flow). Evaluation of cardiovascular and peripheral stents. Inhalation delivery systems such as propellant-metered dose inhalers, and dry powder inhalers (drug dispersion and flow behavior). (10 Hours)
- **Blood Flow and Pressure Measurement Technique:** Mechanics of Sphygmomanometer and Stethoscope. Heart sound and cardiac auscultation. (4 Hours)

### Text books:

1. Wilmer W. Nichols and Michael F. O'Rourke, **McDonald's Blood Flow in Arteries: Theoretical, Experimental and Clinical Principles**, Hodder Arnold Publication
2. Lee Waite, **Biofluid Mechanics in Cardiovascular Systems**, McGraw-Hill
3. Goyal , Megh R, **Biofluid Dynamics of Human body systems**, CRC Press, 2013.
4. C. Kleinstreuer, **Biofluid Dynamics: Principles and Applications**, CRC Press, 2006
5. Ali Ostadfar, **Biofluid Mechanics: Principles and Applications**, Elsevier, 2016



**References:**

1. Biofluid Mechanics 535.661, Johns Hopkins University, School of Engineering
2. Biofluid Dynamics MEC433, UBC College of Engineering
3. Bio Fluid Mechanics (MTR003), Imperial College London
4. BEE 3310 Bio-Fluid Mechanics, Cornell SCE BMED/ME 4757
5. Biofluid Mechanics, Georgia Institute of Technology